

Suggested Teaching Guideline for
Fundamentals of Computer Networks
PGCP-ITISS February 2026

Duration: 40 Theory hrs + 40 Lab hrs + 40 Self-Learning hrs (Total :120 hrs)

Objective: To introduce the student to fundamentals of computers and networks

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Courseware: Data Communications and Networking, Behrouz A. Forouzan, McGraw Hill, 6th Edition

Session 1 & 2: (4T + 4L+ 4SL)

Theory:

- Internetworking
- OSI model
- Ethernet
- Wireless Networking

Lab Assignment:

- Difference between UTP & STP
- Write categories of cables
- e in CAT5e

Self - Learning:

- Write a short summary on the history and evolution of internetworking.
- Research the OSI model and describe how each layer operates in detail. Include examples of real-world technologies or protocols that operate at each layer.
- Go through videos explaining real-world examples of network communication based on the OSI model.
- Look at a real-world networking scenario (e.g., large-scale corporate networks or cloud networks) and analyze how the OSI model applies.
- Study IEEE 802.11 standards (Wi-Fi) and the different frequency bands used in wireless communication.

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Fundamentals of Computer Networks
PGCP-ITISS February 2026

Session 3:(2T + 2L+ 2SL)

Theory:

- Internet Protocol
- TCP/IP model

Lab Assignment:

- IP
- TCP/IP Model
- Write a difference between TCP & UDP.

Self - Learning:

- Write about the key differences between the OSI model and the TCP/IP model.
- Research how TCP/IP was developed and its significance in modern networking.
- Look at a real-life case of an Internet service provider and how TCP/IP is implemented in their infrastructure.
- Why video streaming uses UDP instead of TCP.

Session 4 & 5:(4T + 4L+ 4SL)

Theory:

- IP Subnetting & variable Length Subnet Masking

Lab Assignment:

- Subnetting
- Determine the network and host part of 192.168.5.85 /24, 10.128.240.50/30 address.
- Divide the network as per following requirement 192.168.1.0/24
 - a) Subnet 1 =28 hosts
 - b) Subnet 2 =52 hosts
 - c) Subnet 3 =15 hosts
 - d) Subnet 4 = 5 hosts

Self- Learning:

- Study IPv4 subnetting tools and practice on platforms like SubnettingPractice.com
- Explore how IPv6 is being adopted in modern networking environments (look into IPv6 adoption by major ISPs or enterprises.
- Investigate a global company's transition from IPv4 to IPv6 and the challenges they faced.

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Fundamentals of Computer Networks
PGCP-ITISS February 2026

Session 6:(2T + 2L+ 2SL)**Theory:**

- Router IOS & Security Device Manager

Lab Assignment:

- Subnet the Class C IP Address 195.1.1.0 So that you have 10 subnets each with a maximum 12 hosts on each subnet. List the Address on host 1 on subnet 0,1,2,3,10.
- Divide the network to find 500 hosts in each subnet 152.152.0.0 /16

Self - Learning:

- List router boot sequence steps.
- Explain IOS image storage locations.
- Router misconfiguration leading to network outage.

Session 7:(2T + 2L+ 2SL)**Theory:**

- Managing an Internetworking Router

Lab Assignments:

- Working with Router Booting, configuration registers, Router IOS, Telnet, Resolving Hostname, Debugging.

Self - Learning:

- Find out about Telnet vs SSH
- Find the Importance of logging and debugging
- Explore about Unauthorized Telnet access incident analysis.

Session 8:(2T + 2L+ 2SL)**Theory:**

- Static Routing
- Dynamic Routing
- Routing Protocols

Lab Assignment:

- Dynamic Routing
- Static Routing
- Routing Protocols

Self - Learning:

- Find the difference between Static vs Dynamic routing
- Find the difference between Distance Vector vs Link State routing
- Explore about Administrative Distance Concept
- Explore about Static routing failure in large networks

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Fundamentals of Computer Networks
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Session 9 & 10:(4T + 4L+ 4SL)**Theory:**

- Implementing of Routing Protocols

Lab Assignments:

- Implementation of Static Routing, RIP, IGRP, EIGRP, OSPF

Self - Learning:

- Compare RIP, OSPF, EIGRP
- Find out the difference between IGP vs EGP
- How OSPF used in ISP backbone Networks
- Explore about Routing convergence in enterprise networks

Session 11:(2T + 2L+ 2SL)**Theory:**

- Layer 2 switching
- Spanning Tree Protocol (STP)
- Write types of Spanning Tree Protocol (STP).
- priority number of Spanning Tree Protocol (STP)

Lab Assignments:

- Configuration of switch, STP
- Create the following topology for STP and configure a specific switch as root bridge by changing priority number.
- disable Spanning Tree Protocol (STP)

Self - Learning:

- Find out how STP root bridge election process is done.
- What is Port states in STP
- Loop problems in Layer 2 networks
- Find out the difference between RSTP vs STP
- Explore about Broadcast storm incident due to STP failure.

Session 12:(2T + 2L+ 2SL)**Theory:**

- Virtual LANs
- Inter-VLAN

Suggested Teaching Guideline for
Fundamentals of Computer Networks
PGCP-ITISS February 2026

Lab Assignments:

- Configure, verify, and troubleshoot VLANs (normal/extended range) spanning multiple switches, inter-VLAN routing
- Create following topology where Switch 1 is Server mode and switch 2,3,4,5 are client mode. Use network 192.168.10.0 /24
 - Transfer the Vlan info from server to all clients.
 - Assign vlan as mention in the topology.
 - PC's in having same vlan should ping each other.
- Create following topology where Switch 1 is Server mode and switch 2,4, are client mode.
Use network 192.168.1.0 /24
 - Transfer the Vlan info from server to all clients.
 - Assign vlan as mention in the topology.
 - PC's in having same vlan should ping each other.
 - Configured switch 3 in transfer mode and transfer vlan info to switch 4

Self - Learning:

- Learn about VLAN tagging and trunking concepts
- Explain the Inter-VLAN routing
- Find out about VTP modes and risks
- How VLAN's in data centers is used
- VLAN segmentation for HR, Finance, and IT departments

Session 13:(2T + 2L+ 2SL)**Theory:**

- Infrastructure Security
- ACL
- TACACS+
- RADIUS

Lab Assignments:

- Configure, verify, and troubleshoot port security, describe common access layer threat mitigation, Configure, verify, and troubleshoot IPv4 and IPv6 access list for traffic filtering techniques,
- Verify ACLs using the APIC-EM Path Trace ACL analysis tool
- Configure, verify, and troubleshoot basic device hardening
- Describe device security using AAA with TACACS+ and RADIUS

Self - Learning:

- Find out about ACL types and its use cases
- Port security violation modes

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Fundamentals of Computer Networks
PGCP-ITISS February 2026

- Find the difference between TACACS+ vs RADIUS
- Explore about how Insider attack prevented using ACL's and AAA

Session 14:(2T + 2L+ 2SL)

Theory:

- NAT
- IPV6
- WAN Technologies

Lab Assignments

- NAT configuration, Configuring Routers with IPv6

Self - Learning:

- Write about the types of NAT with examples
- Find the difference between IPV4 vs IPV6 comparison
- Find out about the IPV6 adoption challenges
- How WAN technologies used by ISP's
- Explore about NAT usage in home routers

Session 15:(2T + 2L+ 2SL)

Theory:

- PPP & MLPPP
- PPPoE

Lab Assignments:

- Configure and verify PPP and MLPPP on WAN interfaces using local authentication
- Configure, verify, and troubleshoot PPPoE client-side interfaces using local authentication, Configure, verify, and troubleshoot GRE tunnel connectivity.
- Configure and verify single-homed branch connectivity using eBGP IPv4 (limited to peering and route advertisement using Network command only)

Self - Learning:

- Learn about PPP authentication Mechanisms
- Learn about GRE tunneling
- Difference between MPLS vs traditional WAN
- Learn about VPN fundamentals
- Explore about Secure branch connectivity using PPP and GRE

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Session 16:(2T + 2L+ 2SL)**Theory:**

- Introduction to SDN,
- Overview and Architecture of SDN
- Scalability (Data Centers, Service provider networks, ISP Automation)
- Reliability (QoS, and Service Availability)
- Consistency (Configuration management, and Access Control Violations)
- Opportunities and Challenges

Lab Assignment:

- SDN and Architecture of SDN.
- Control plane vs data plane

Self - Learning:

- How SDN use in cloud data centers
- Find the difference between SDN vs traditional networking
- Explore about Google's SDN0- based network (B4)

Session 17:(2T + 2L+ 2SL)**Theory:**

- Virtual networking
- Use-cases (Network Access Control, Virtual Customer Edge, Datacenter Optimization)

Lab Assignment:

- Virtual networking
- Use cases of Network Access Control, Virtual Customer Edge, Datacenter Optimization

Self - Learning:

- Find the difference between Virtual switch vs physical switch
- Learn about Network virtualization benefits
- Find about NFV and SDN relationship

Session 18:(2T + 2L+ 2SL)**Theory:**

- Introduction to OpenFlow
- History and evolution
- Control and data plane separation
- OpenDaylight architecture overview

Lab Assignment:

- Understanding the classroom environment
- Installing OpenDaylight
- Initial Configuration

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- Initial Controller Interactions

Self - Learning:

- Find about OpenFlow messages types
- Find about Controller- switch interaction flow
- Find about Openflow versions evolution
- SDN controllers comparison
- OpenFlow in academic research networks.

Session 19:(2T + 2L+ 2SL)

Theory:

- Getting Started with Opendaylight
- How to get started
- OpenDaylight clustering
- Model-Driven Service Abstraction Layer
- Internal datastore
- OpenFlow plugin
- OpenVSwitch concepts
- Mininet overview
- L2Switch Application

Lab Assignment:

- Configuring an OpenDaylight Cluster
- Initial Command Line Interactions
- Enabling Required Features
- Launching the User Interface
- Creating an Emulated SDN Network
- Running L2Switch

Self - Learning:

- Find about MD-SAL components
- Learn about Open vSwitch architecture
- Find the difference between Emulated vs real SDN networks
- Explore about L2 Switch application behavior analysis

Session 20:(2T + 2L+ 2SL)

Theory:

- OpenDaylight and AAA
- Introduction to OVSDb Virtualization
- Application Intents and Group Based Policy
- Service Function Chaining
- LISP Flow Mapping
- Virtual Tenant Networks

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Lab Assignment:

- Advanced AAA integrations
- Creating Virtual Networks
- Working with policies and intents
- Initial Service Function Chaining
- OpenDaylight and OpenVSwitch

Self - Learning:

- Find out about Group based policy
- Learn about SDN Security challenges
- Find out about Multi-tenant virtual networks
- Explore about Virtual tenant isolation in data centers